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DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Presentation Outline

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Problem Statement

Problems faced as a Student

- In most cases, a student has to navigate through numerous application forms, websites, portals, and documents just to gather information about his/her academic requirements.
- It makes the process of tracking events quite difficult and the user may easily overlook crucial information or deadlines.
- Another challenge that students face when working online with organizations is that it may sometimes be rather difficult to distinguish an email message from an official sender from an ordinary spam email.
- Campus Copilot and Smart Spam Detector will design a revolutionary and integrated information system wherein the user will easily access all the necessary information about campus avoiding phishing emails.

Problem Understanding

1. **Who Experiences This Problem?**
   - **Students & Freshers:** Difficulty navigating multiple campus portals.
   - **Faculty & Administration:** Repeated student queries.
   - **IT Support Teams:** Platform access issues and email security concerns.

2. **What Causes the Problem?**
   - **Data Fragmentation:** Information is scattered across websites, apps, PDFs, and emails.
   - **Limited Integration:** Campus systems operate independently.
   - **Lack of AI Assistance:** Students must manually search for information.
   - **Phishing Threats:** Fraudulent emails use deceptive links and spoofed identities.

3. **How Does It Affect Users?**
   - **Time Loss:** Academic information takes longer to find.
   - **Missed Updates:** Important announcements and deadlines go unnoticed.
   - **Security Risks:** Phishing can expose credentials and sensitive data.
   - **Increased Workload:** Repeated queries burden institutional support teams.

Existing Gaps & Challenges

1. **Fragmented Campus Information**
   - Academic information is scattered across multiple websites, apps, PDFs, and communication channels, making it difficult for students to access everything in one place.

2. **Limited Campus-Specific AI Assistance**
   - Existing portals offer limited conversational support and lack integrated, context-aware retrieval of institutional information.

3. **Email Threat Detection Limitations**
   - Conventional email filters may miss targeted phishing emails, making it difficult to distinguish genuine college communications from potentially fraudulent messages.

Our Proposed Solution

1. **Core Idea or Product**
   - An AI-powered unified platform combining campus information assistance with smart email threat detection.

2. **How the Solution Works**
   - **Campus Copilot** uses RAG to retrieve authorised academic information and answer student queries.
   - **Spam Detector** analyses sender details, suspicious links, and message content to flag potential threats.

3. **Main Value Delivered**
   - Provides simplified information access, personalised assistance, and improved email security awareness through a single student-focused platform.

Technical Approach

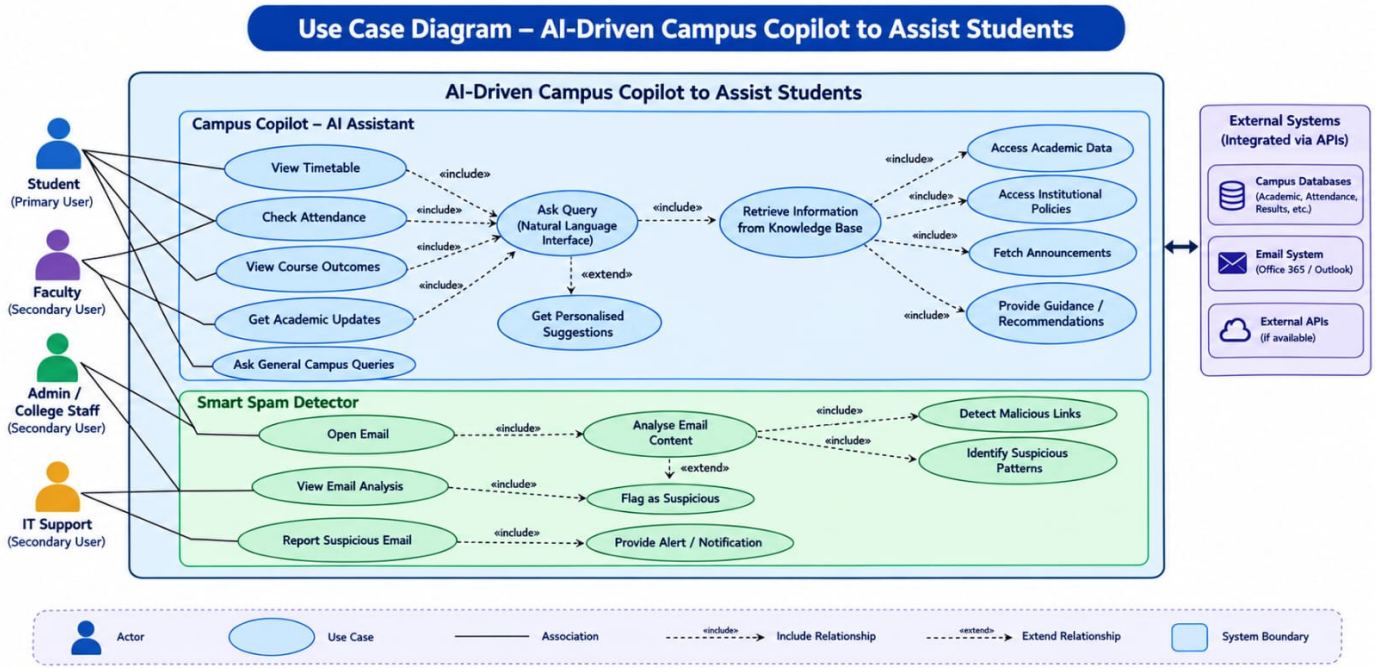


Figure: Flowchart of Operation of the Program

Tech Stack

Frontend: Tailwind CSS, Flutter (mobile app), React.js (webpage)

Backend: Node.js, Python, REST APIs

Database: MongoDB (unstructured data), PostgreSQL (structured data)

AI/ML, APIs: OpenAI/Llama (LLM), college or university APIs, authentication

Hardware/Infrastructure: Cloud hosting (AWS), secure storage and monitoring, load balancers and firewalls

Innovation & Uniqueness

Innovation

Combines conversational AI for campus information retrieval with intelligent email threat detection in one platform.

Problem–Solution Fit

Uses institution-specific knowledge retrieval and suspicious-email analysis to address fragmented information access and phishing risks.

Key Differentiator

Integrates academic assistance and email security into a unified, student-focused platform.

Feasibility & Scalability

1. **Feasibility**
   - Develop using Python, AI/ML tools, a database, and a web/mobile framework.
   - Key steps include UI development, campus data integration, RAG implementation, and email threat detection.

2. **Scalability**
   - Start with essential academic services.
   - Expand to more departments and campus systems using modular architecture and APIs.

3. **Risks & Mitigation**
   - **Risks:** Data access, privacy, inaccurate responses, and false spam alerts.
   - **Mitigation:** Authentication, access control, secure data handling, verified sources, and continuous testing.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases

1. **Campus Copilot**
   - **User:** College Student
   - **Scenario:** A student uses Campus Copilot to find important campus information, such as class schedules, classroom locations, faculty details, assignment deadlines, and college announcements. The system provides quick and organized answers, helping the student save time and avoid confusion.

2. **Spam Detector**
   - **User:** College Student
   - **Scenario:** A student receives suspicious messages, emails, or links related to college activities. The Spam Detector analyzes the content and identifies potentially unwanted or fraudulent messages, helping the student avoid scams and unwanted communication.

3. **Integrated Campus Copilot and Spam Detector**
   - **User:** College Student
   - **Scenario:** A student uses Campus Copilot to access campus information and receive important notifications. The integrated Spam Detector filters suspicious messages while allowing genuine campus updates to reach the student. This makes communication more organized, convenient, and secure.

Impacts and Benefits

1. Benefits to Target Users

- **Students:** Quick campus information and spam alerts.
- **Faculty:** Reduced repetitive queries.
- **IT teams:** Improved email security.

2. Impacts

- **Social:** Safer communication.
- **Economic:** Saves time and costs.
- **Environmental:** Reduces paper usage.

3. Measurable Outcomes

- Faster information access.
- Better spam detection.
- Fewer repetitive queries.
- Reduced paper usage.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.3	Quality Education	Makes campus information easily accessible to students.
SDG 9	Target 9.c	Industry, Innovation and Infrastructure	Promotes AI-based digital campus services.
SDG 16	Target 16.6	Peace, Justice and Strong Institutions	Supports safer digital communication and phishing awareness.
SDG 12	Target 12.5	Responsible Consumption and Production	Reduces paper usage through digital information sharing.

Main SDGs: SDG 4, SDG 9, and SDG 16.

Individual Member Contribution

Member	Contribution or responsibility
Jeeva Rakshitha M	Collecting Data From Research Papers
Pradeesh Kanna M	Integrating works of team members into a unified system
Dharshan S	Finding Tech Stacks to solve the real world Problem
Thaarigaa D	Summarize the collected Data into desirable content

Yukti ID

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